(a) Input

prepared by the user, before the run

Group A
10 raw files

Group B
10 raw files

Sample	Group
P173	A
P176	A
⋮	⋮
P528	B

sample_Group.csv
one row per file, Group is A or B

what one sample becomes

Sample	RT	RT&m/z	m/z	Int.	blank
173	2	2101.023	101.0234	3.2e5	

one block per sample · one blank column between blocks
this layout is the only contract between the two tools

(b) MSGeter

tab 1 of the app

Ms1Geter — primary

in .raw / .mzML / .mzXML + sample_Group.csv

1-min bins → pseudo-spectrum → 10 ppm → I > 10,000 → RT = bin + 1

out

- Clean_MS1_Data_Group{A,B}.xlsx
  six-column blocks, one per sample, 18 one-minute bins
- MS2_Group_Info.csv
  every MS2 scan: scan, RT, precursor m/z, group
- MS2_MGF/<sample>.msp
  one block per MS2 spectrum

no EIC and no peak integration — only the minute bins and their spectral peaks

Ms2Geter — optional

MS2_Group_Info.csv + feature list

↓

targets-only .msp, one per feature

matches |precursor − m/z| ≤ 0.01 Da, then the closest RT

RtGeter — optional

peak table (id, sample, m/z, RT_min) + raw folder

↓

the same table with one RT column added

RT_min is the right edge: 2 means the window (1, 2]

(c) DMetFinder

tab 2 of the app

Clean_MS1_Data_GroupA.xlsx
Clean_MS1_Data_GroupB.xlsx

two workbooks, one per group

1. **read** — 6-col blocks → rtMz
2. **matrix** — ±0.002, count & avg
3. **50% / 80%** — floor(0.5n) / ceil(0.8n)
4. **score** — intensity × count × rel
5. **isotope** — M+1 / M+2 / M+3
6. **A vs B** — ±0.003, abundance ratio

out

- A组-筛选50%和80%离子.xlsx 3
- A组-离子打分.xlsx 3
- A组-同位素离子查找.xlsx 1
- A组-80%离子与B组总离子比对分析.xlsx 1
- **A组-B组差异离子筛选汇总.xlsx** 2

9 workbooks / 18 sheets in total
the B-side files mirror these four, minus the summary

A组和B组差异离子(A组比B组)

m/z	count A/B	avg A/B	FC	p	M+1
267.1231	9 / 0	19,203 / 0	–	0.0031	**0**
214.0864	8 / 8	42,981 / 22,614	1.9	0.0074	0
219.0899	8 / 8	38,112 / 18,143	2.1	0.0092	1
301.1077	7 / 7	25,470 / 31,838	0.8	0.0410	0
336.1590	5 / 5	14,905 / 4,384	3.4	0.0055	0

p is a live Excel formula =TTEST(...); M+1 = 1 tags an isotope, it is not removed

(d) Downstream

differential ion list
+ MS2 spectra (.mgf / .msp)

- **SIRIUS** — molecular formula
- **MetGem** — molecular network
- **EntropySearch** — spectral entropy
- **MS-DIAL** — library match

structure, class and network for the differential ions